Data Cleaning, Transformation and Modelling

Theoretical Questions

Question 1 : What is Data Loading in Power BI and why is it considered the first step of analysis?

Answer:

Data Loading in Power BI refers to the process of connecting to external data sources (such as Excel files, SQL databases, CSV files, or cloud services), importing that data, and bringing it into the Power BI ecosystem (Power Query or the data model) for exploration and reporting.

It is considered the first step of analysis because:

- **Foundation of Insights:** No analysis or visualization can take place without raw data. Loading acts as the bridge between source systems and the reporting interface.
- **Data Discovery & Schema Recognition:** During loading, Power BI scans the data structures, detects column types, and uncovers initial shapes, laying the groundwork for proper structuring.
- **Prerequisite for Transformation and Modeling:** Before you can clean, transform, or build relationships, the data must first be successfully ingested into the workspace.

Question 2 : Explain the difference between “Load” and “Transform Data” in Power BI.

Answer:

- **Load:** The "Load" option brings the data directly from the source into the Power BI data model without making any structural changes. It executes the query and saves the tables into memory so you can immediately begin creating visuals and relationships.
- **Transform Data:** The "Transform Data" option opens the **Power Query Editor** before the data is loaded into the model. This allows you to clean, shape, and modify the data (e.g., removing duplicates, handling null values, renaming columns, changing data types, merging or appending queries). This ensures that only clean, optimized data enters your model, improving performance and accuracy.

Question 3 : What is a Fact Table and a Dimension Table? Give examples from the dataset.

Answer:

- **Fact Table:** A fact table stores quantitative, numerical measurements or metrics resulting from a business process or event (e.g., sales amounts, transaction counts, or in the context of this dataset, daily recorded metrics like confirmed cases, deaths, and recoveries). It typically contains foreign keys linking to dimension tables and has many rows.
    - *Example from dataset:* A table containing numerical metrics such as `Confirmed`, `Deaths`, and `Recovered` against specific dates and geographic regions.
- **Dimension Table:** A dimension table stores descriptive attributes and context surrounding the business process (e.g., customer details, product categories, locations, or dates). It provides context to the numbers in the fact table and usually has fewer rows with unique identifiers (primary keys).
    - *Example from dataset:* A Country/Region lookup table containing columns like `Country`, `State/Province`, and region-specific metadata.

Question 4 : Why is Star Schema preferred over Snowflake Schema in Power BI?

Answer:

A Star Schema is preferred over a Snowflake Schema in Power BI for several reasons:

- **Simpler Relationships:** In a star schema, dimension tables are directly connected to the central fact table in a denormalized manner, avoiding multi-level hierarchies.
- **Better Performance:** Fewer joins are required during query evaluation, which optimizes VertiPaq engine performance, speeds up report rendering, and reduces processing overhead.
- **Easier Usability:** Business users and report creators find star schemas much more intuitive to understand and navigate when building DAX measures and exploring fields.

Practical Questions

Question 5 : Identify and remove duplicate records based on Date, Country, and State.

Answer:

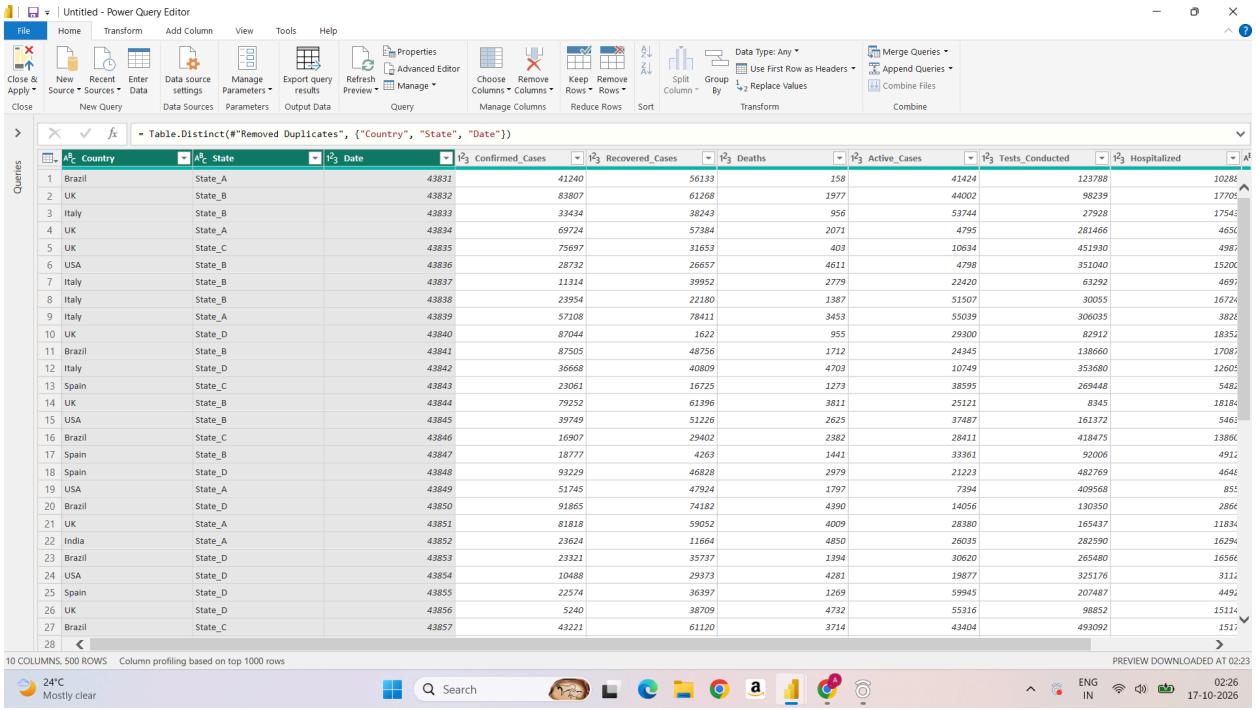
To remove duplicates based on specific columns in Power BI (Power Query):

1. Open **Transform Data** to launch the **Power Query Editor**.
2. Select the columns: `Date`, `Country` (or `Country/Region`), and `State` (or `Province/State`).
3. Right-click on the header of one of the selected columns (or go to the **Home** tab ribbon).
4. Click on **Remove Rows** -> **Remove Duplicates**.
5. Alternatively, you can use Group By or reference the M-code step:
6. Code snippet

= Table.Distinct(#"Previous Step", {"Date", "Country", "State"})

7.

Practical proof :



Question 6 : Identify and replace null values in `Vaccination_Status`.

Answer:

To handle null values in the `Vaccination_Status` column using Power Query:

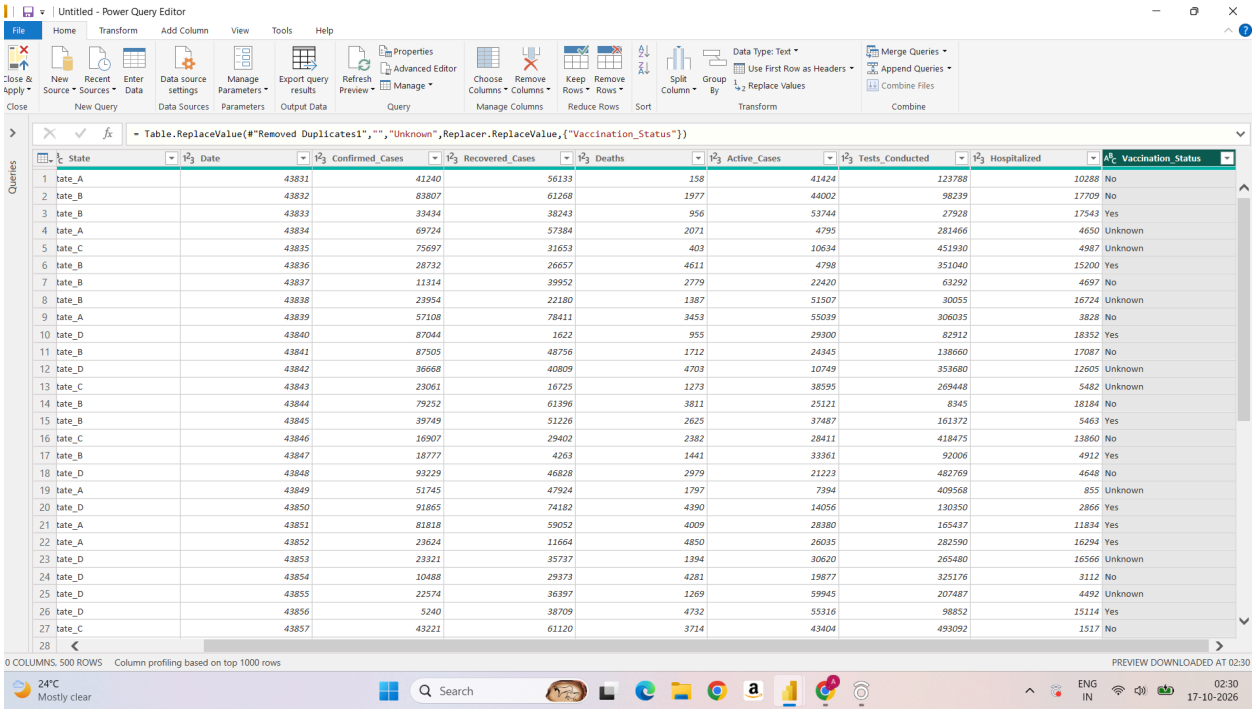
1. In the **Power Query Editor**, locate the `Vaccination_Status` column.
2. Select the column, right-click the header, and select **Replace Values...** (or go to the **Transform** tab and click **Replace Values**).
3. In the "Value To Find" field, leave it blank (if handling true nulls) or type `null` if represented as text, and enter your replacement value (e.g., `"Unknown"` or `"Not Available"`) in the "Replace With" field.
4. Alternatively, you can use the right-click menu option **Replace Errors / Replace Nulls** or use the M-code step:
5. Code snippet

= Table.ReplaceValue(#"Previous Step", null, "Unknown", Replacer.ReplaceValue, {"Vaccination_Status"})

6.

7.

Practical proof :



Question 7 : Create a new column to calculate Recovery Rate.

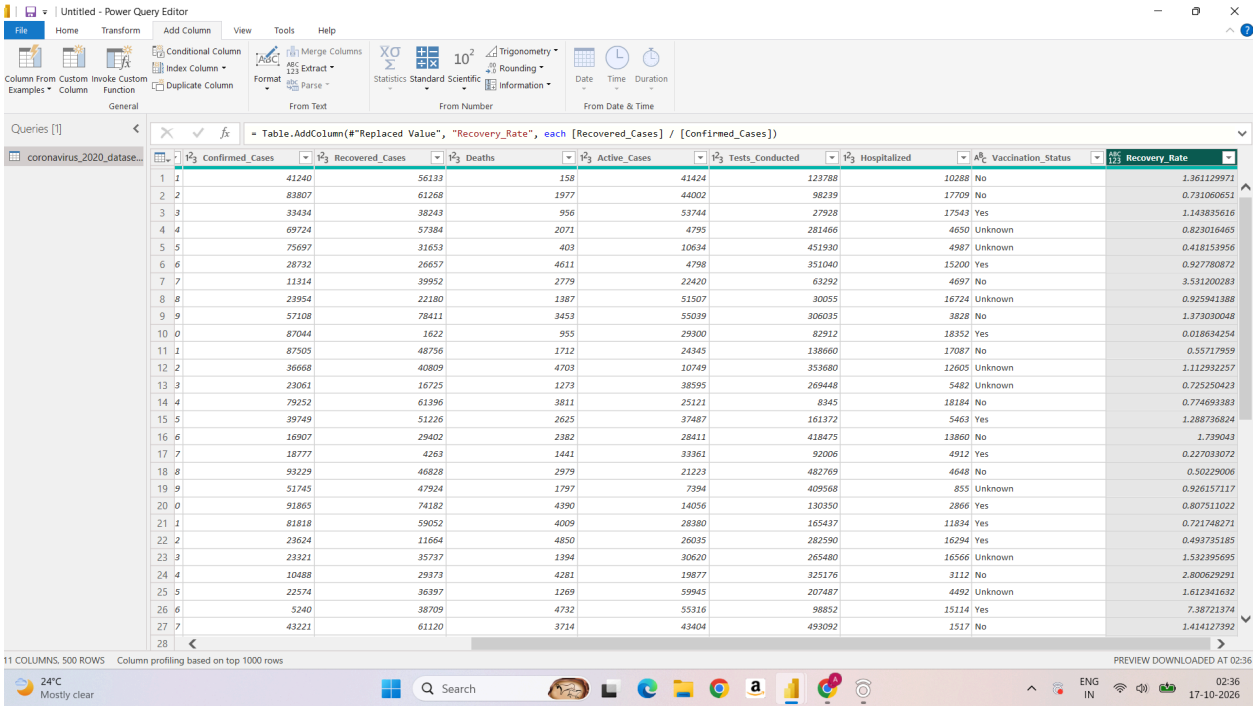
Answer:

To calculate the Recovery Rate as a calculated column using DAX:

Code snippet

```
Recovery Rate =
IF(
    [Confirmed] > 0,
    DIVIDE([Recovered], [Confirmed]),
    0
)
```

Practical proof :



Question 8 : Create a summarized table showing total confirmed cases by Country.

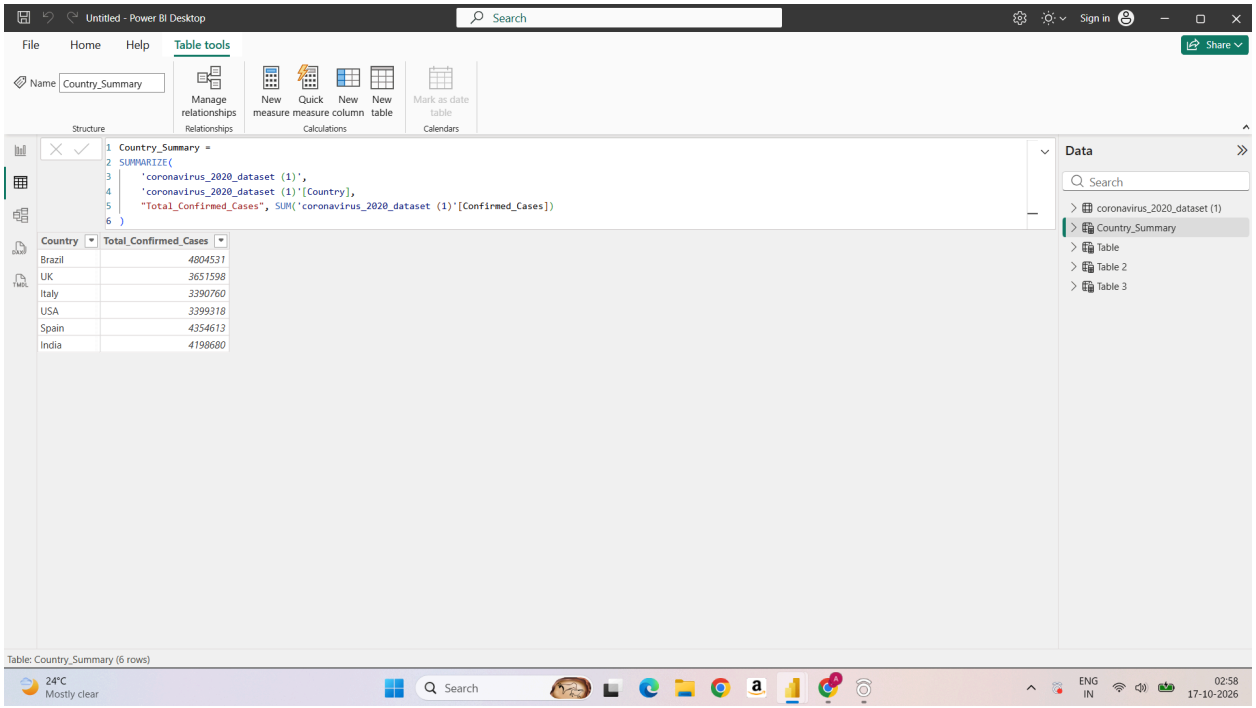
Answer:

To create a summarized table using DAX (via **New Table** under Modeling):

Code snippet

```
Country_Summary =
SUMMARIZE(
    Corona_Virus_2020,
    Corona_Virus_2020[Country],
    "Total_Confirmed_Cases", SUM(Corona_Virus_2020[Confirmed])
)
```

Practical proof :


Untitled - Power BI Desktop
Search
Sign in
File
Home
Help
Table tools
Share
Name
Country_Summary
Manage relationships
New measure
Quick measure
New column
New table
Mark as date table
Structure
Relationships
Calculations
Calendars
1 Country_Summary =
2 SUMMARIZE(
3 'coronavirus_2020_dataset (1)',
4 'coronavirus_2020_dataset (1)'[Country],
5 "Total_Confirmed_Cases", SUM('coronavirus_2020_dataset (1)'[Confirmed_Cases])
6)
Country
Total_Confirmed_Cases
Brazil 4804531
UK 3651598
Italy 3390760
USA 3399318
Spain 4354613
India 4198680
Data
Search
coronavirus_2020_dataset (1)
Country_Summary
Table
Table 2
Table 3
Table: Country_Summary (6 rows)
24°C
Mostly clear
Search
ENG
IN
02:58
17-10-2026